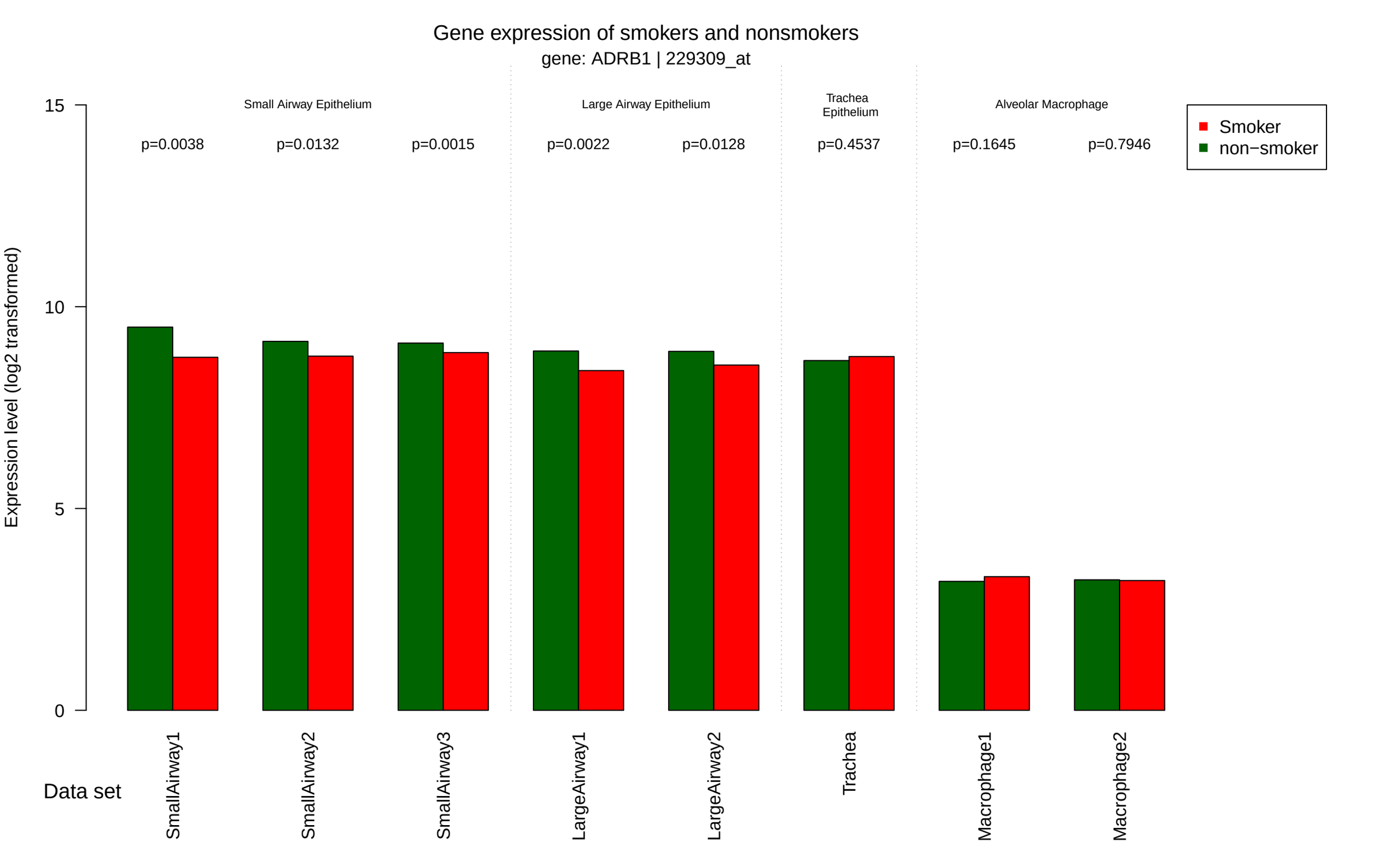
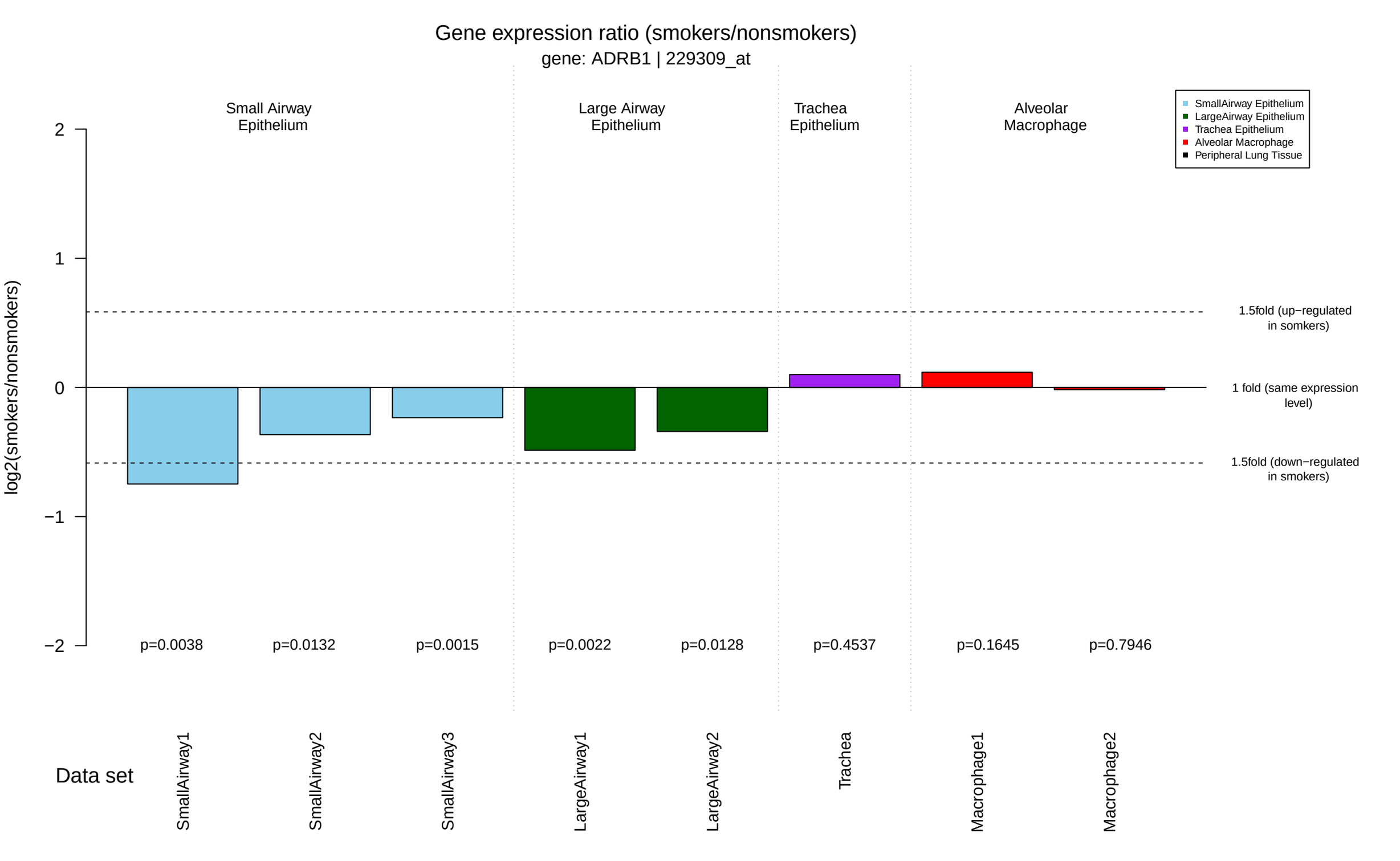
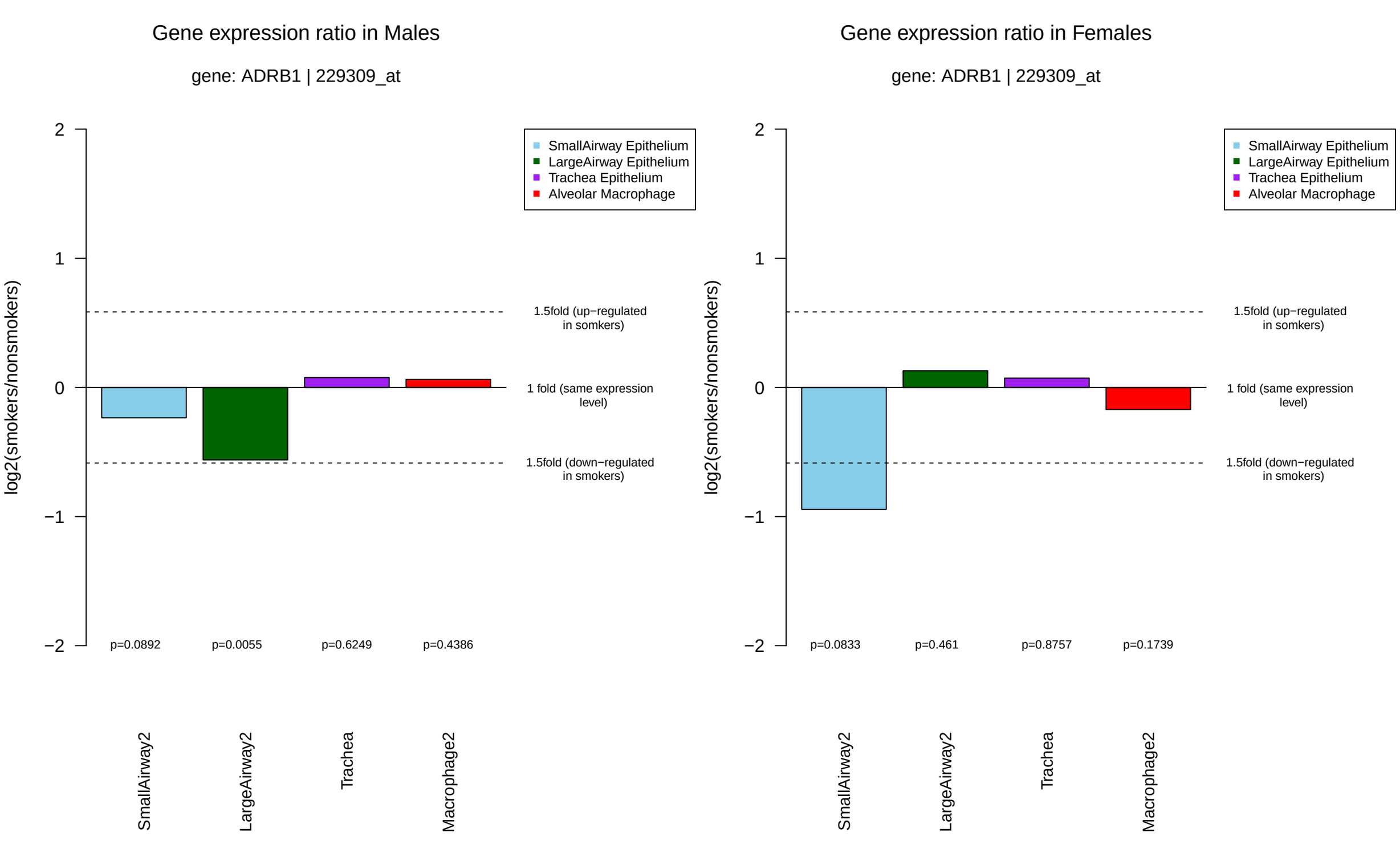
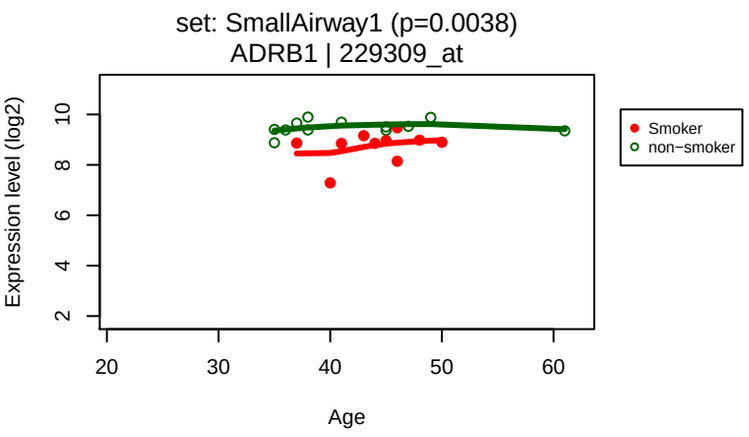
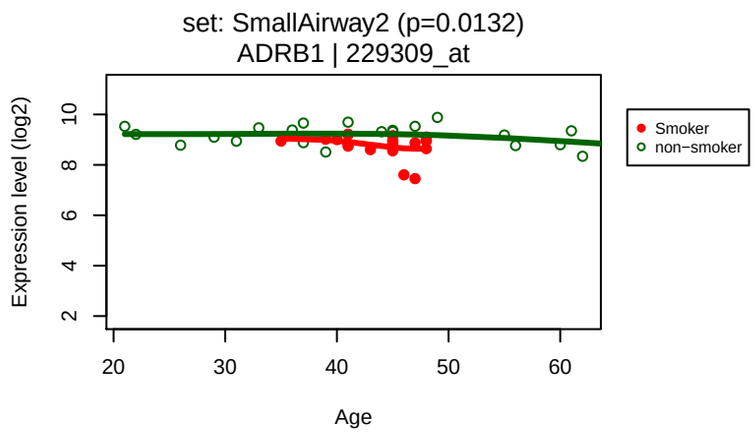
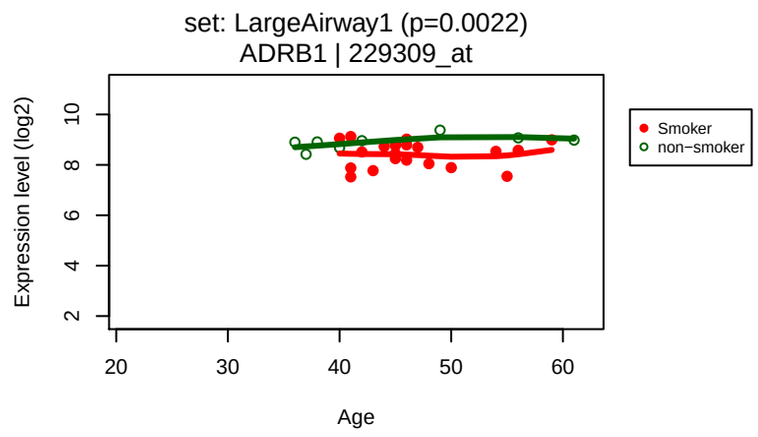

Gene expression of smokers and nonsmokers
gene: ADRB1 | 229309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0038
p=0.0132
p=0.0015
p=0.0022
p=0.0128
p=0.4537
p=0.1645
p=0.7946
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

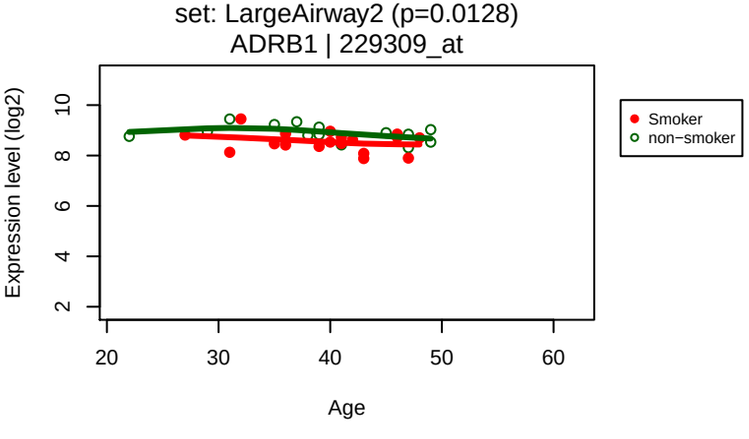

Gene expression ratio (smokers/nonsmokers)
gene: ADRB1 | 229309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0038
p=0.0132
p=0.0015
p=0.0022
p=0.0128
p=0.4537
p=0.1645
p=0.7946
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

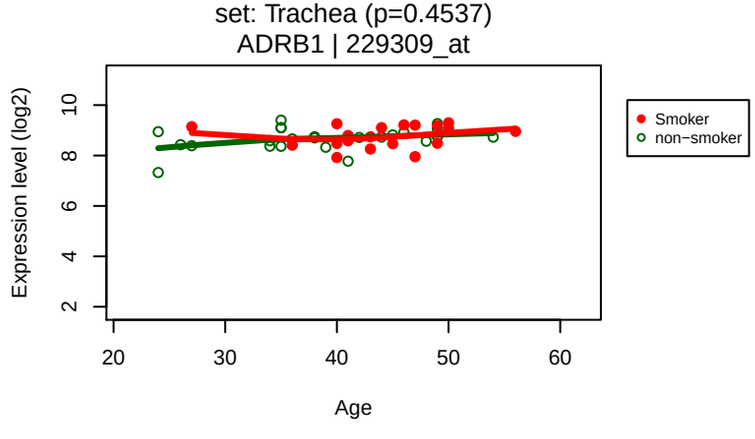

Gene expression ratio in Males
gene: ADRB1 | 229309_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0892
p=0.0055
p=0.6249
p=0.4386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRB1 | 229309_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0833
p=0.461
p=0.8757
p=0.1739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

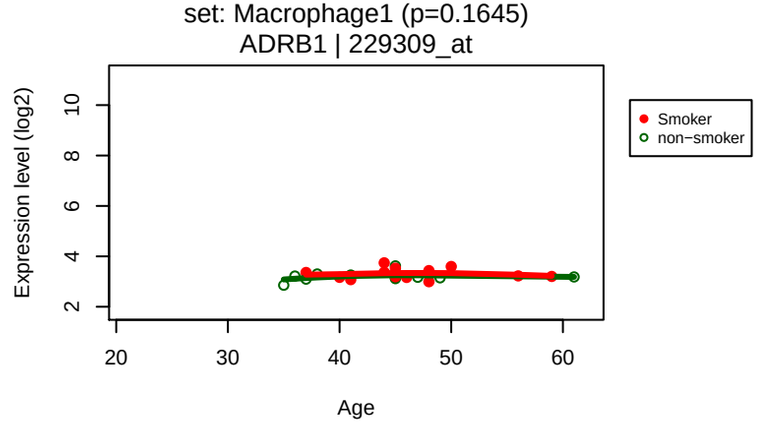

set: SmallAirway1 (p=0.0038)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker

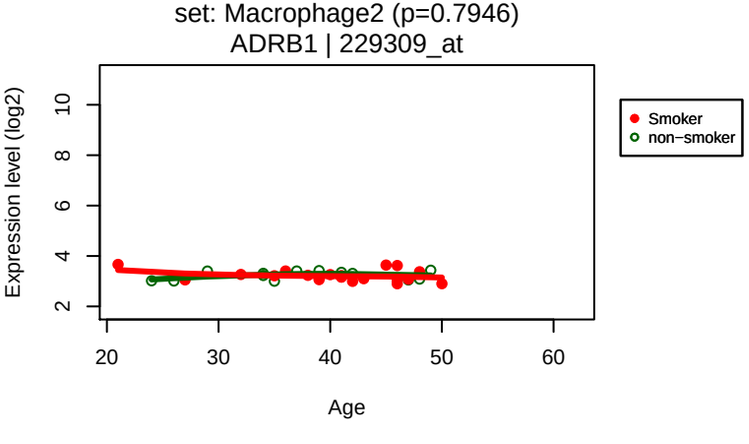

set: SmallAirway2 (p=0.0132)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker

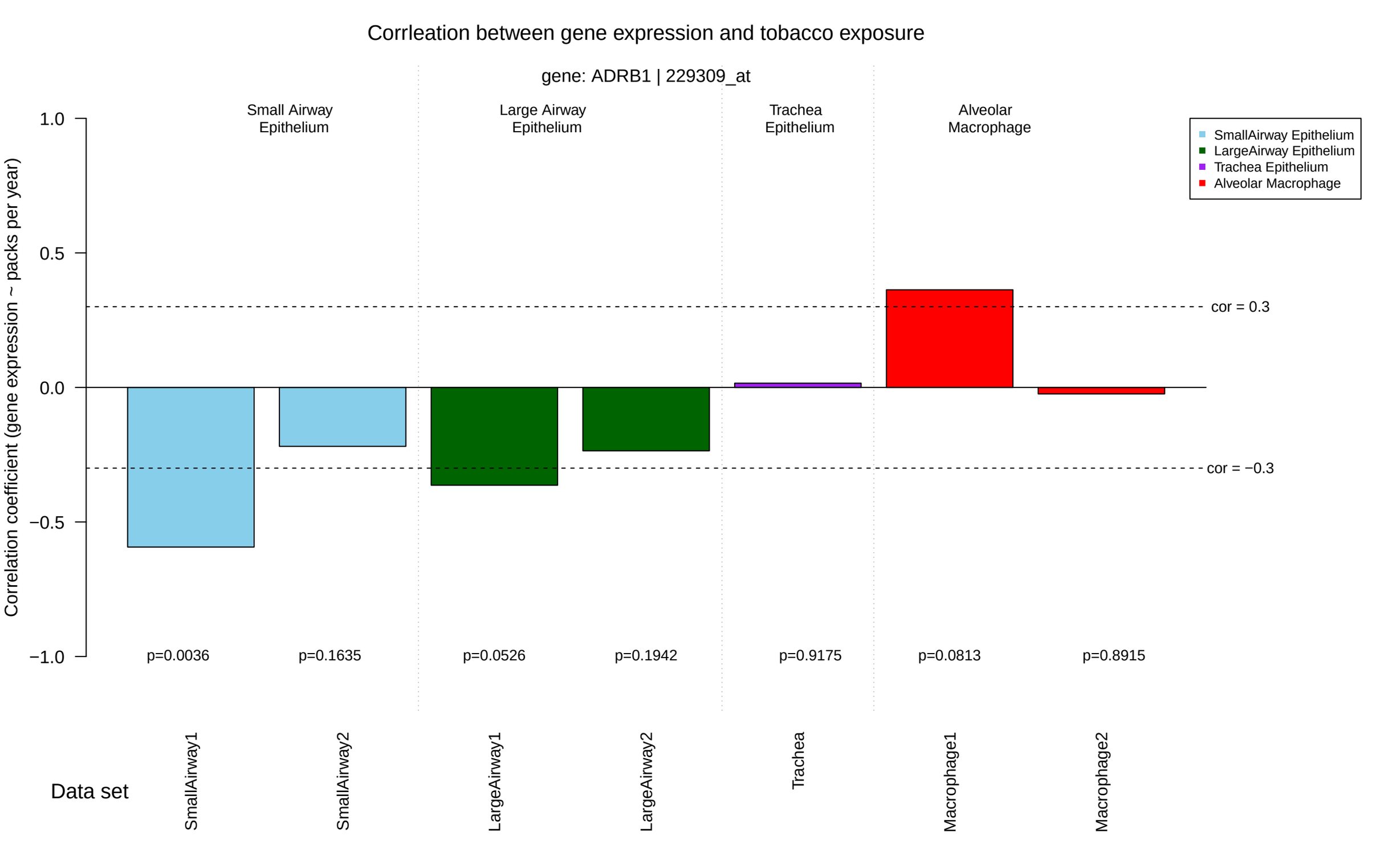

set: LargeAirway1 (p=0.0022)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0128)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4537)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1645)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7946)
ADRB1 | 229309_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRB1 | 229309_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0036
p=0.1635
p=0.0526
p=0.1942
p=0.9175
p=0.0813
p=0.8915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2